
PROFILE

Hard working, team oriented, self-motivated applications engineer and systems integrator with a passion for delivering on site tech enabled services to the highest industrial standards. A proven track record across seven years designing, costing, deploying, configuring, and integrating RFID and RTLS systems on Fortune 100 and federal sites. Comfortable in all RF project environments and roles including as a project presales engineer, design consultant, delivery engineer, project manager, software configuration and integration expert, and in optimization and support of live systems. Committed to integrity and excellence. Driven by a desire to make a meaningful difference through the application of technology and innovation.

EDUCATION

COLORADO SCHOOL OF MINES – Golden, Colorado

Bachelor of Science – May 2017

Varsity Football
Engineering

Major: Mechanical

FLATIRON SCHOOL – Denver, Colorado

Software Engineering Bootcamp – August 2023

SKILLS & STRENGTHS

- **RF Subject Matter Expert:** Expert in wide variety of Radio Frequency (RF) technologies including HF RFID, UHF RAIN RFID, BLE, UWB, GPS, RTK GPS. Extensive hardware and software skills developed deploying thousands of devices for leading RF technology providers such as HID, Zebra, Quuppa, RF Controls, Ubisense, Turck, SensThys, etc.
 - **Real-Time Location Systems (RTLS):** Design, demo, planning, delivery, support and performance analysis of HF RFID, UHF RFID, UWB, GPS, RTK GPS, BLE, RFID, LiDAR, and machine vision applications.
 - **Software Development:** Python, JavaScript, HTML, CSS, React, React Native, Excel VBA, DB, and full-stack solutions. Custom Applications Development. Bash/Python Scripting, CAD plugins, Excel-to-software integrations.
 - **Automation Integration:** Custom HMIs and software integrations. XML, OPC tags, PLCs, and tool controller protocols. RF emitting devices, sensors, tags, reader, printers, scanners, ERP, applications support.
 - **Field Survey and Site Work:** Manual total station, GNSS/GPS, RTK GPS, LiDAR, aerial photography, satellite imagery, AutoCAD; CAD-to-SVG web, as-builts, layout verification, site and multi-building coordinate systems, 3D Helmert transforms, inner building local to N/E state-plane (m, ft, in, lat, long), building rotations, angular coordinates pitch/roll/yaw. Site Digitalization.
 - **Project Travel and Management:** Planning, Costing, Delivering complex systems on time and within budget. Weekly travel or multi-week air bnb or hotel, on-site work and remote work. One-person deploys. Facility, factory, construction, site, campus, yard, hangar, and base.
 - **Entrepreneurial Mindset:** Self-starter with a proven ability to build and lead. Equipment procurement, with ties to manufacturers and renters providing project cost savings and flexibility.
 - **Technical Communication:** Skilled in translating complex technical concepts for audiences with a diversity of professional backgrounds.
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WORK EXPERIENCE

Founder | Realtime Engineering and Development LLC (RED) – Windsor, Colorado

February 2024 – Present

Applications Engineer, RF Field Engineer and Surveyor, Industrial / Federal Maritime Contractor. Device and Technology Agnostic RTLS solutions planning, costing, deployment, and integration. Experience applied to logistics,

factory and facility management and automation, and other custom location system use cases significantly improving operational process efficiency, clarity, safety, reducing task time and costs. With a particular focus on empowering businesses to secure large RTLS contracts with dynamic and evolving requirements. Have custom RTLS project planning, pricing, and deployment software application. Analyze client project feasibility and cost for multi-source location systems combining HF RFID, UHF RAIN RFID, UWB, BLE, GPS, RTK GPS, and more. Build scalable software for clients, including custom front-end and back-end solutions. Leverage Python for backend development and React Native for front end development. Provide clients comprehensive integration engineering and custom use case mobile applications development.

SME Level 3 RF Engineer | Federal Maritime Subcontractor – United States**2023 – Present**

RF System Surveyor and Application Configuration Engineer. Surveyed devices locally due to missing drawing files and facilities changes, then built an excel workbook that utilized a 3D Helmert transform to transform local survey coordinates and device rotations (pitch roll, yaw) into a northing/easting state-plane system. Carried network properties and radar configuration with the coordinates in workbook and created custom integration with client's software to import 5000 data points into 4 separate applications, reducing 20,000 manual data entries. Imported device set into CAD and SVG web maps with custom python scripting and developed CAD plugin. Generated as built .dwg layouts and .svgs for the customer, client, and customer mobile application's front end web map display. The resulting Excel-to-software toolkit and integration saved months of manual labor. Contract includes training: taught two on site technicians to run a manual total station and survey devices. Priced project and developed project plans with project lead in presales capacity which was instrumental in landing prime contractor large contracts related to Ships for America Congressional Act. Field phase paused Feb 2026 due to ongoing conflict.

CTO & Lead Developer | MeanderEV – Denver, Colorado**April 2023 – December 2023**

Spearheaded the development of a mobile EV charger-sharing platform. Developed scalable application framework with React Native and Python backend and object relational mapping system. Created application demo for investors. Led application technical roadmap, projecting development costs and timeline. Presented roadmap and application demos to partners and investors.

Applications Engineer & Presales | Ubisense – Cambridge, United Kingdom**August 2019 – November 2022**

Designed and delivered RTLS-based automation solutions to fortune 100 clients such as John Deere, Lockheed Martin, GM, Daimler, Honda, etc. Developed and delivered demos and technical presentations to secure high value contracts. Integrated advanced location hardware with customer systems, optimizing process safety, quality, and efficiency. Assisted in development of an advanced algorithmic calibration system utilizing LiDAR to dramatically reduce RTLS system deployment time and cost.

Commercial PV System Designer & Presales | RGS Energy – Denver, Colorado**August 2017 – January 2019**

Designed residential and commercial photovoltaic systems for installation, including battery storage solutions. Developed commercial PV engineering and costing software to reduce project cycle time. Provided cost analyses and ROI estimates for large commercial ($\geq 250\text{kW}$) renewable energy projects. Managed teams of contractors and in house electricians in +10 states in coordination with construction management. Technical calls to advise utility company management on grid tied batteries as they first became commercially available including Eversource, National Grid, and other smaller utilities.

NOTABLE PROJECTS

Federal Maritime / Industrial Contract**United States**

- Cost RTLS system and assisted prime contractor in writing proposals and landing \$60M contract related to Ships for America Congressional Act.
- Developed costing software to assist in pricing rapid changes in project requirements and scope.
- Supported various prime contractor management roles in consulting role.
- Developed digital assets such as site .CAD files and .SVG Webmaps for a large site utilized in the mobile end application. Digital maps were created from stitched satellite imagery, CAD files, LiDAR scans, and aerial photography.

- Surveyed 450 HF, UHF RAIN RFID, BLE, and RFID devices across 10 buildings in two weeks within 1in using manual total station. Local surveys per building; 3D Helmert into state-plane northing/easting in inches).
- Configured all 450 devices working with 4 different vendors and their respective software and technical support teams (Zebra, Quuppa, RF Controls, SensThys) Remote booting and configuration utilizing mirrored network settings.
- Pitch/roll/yaw on all BLE and UHF RFID devices.
- Excel-to-software integration that saved 20,000 manual data entries.
- On-site travel. Equipment Rental.

John Deere X9 Grizzly Line and EOL RTLS Deployment**Moline, Illinois**

- Designed project. Traveled on site, debrief with JD Factory Department Head, IT Lead, and Automation Team Leader, Walk assembly line, understand tracking use case and requirements, take measurements with hand laser, pictures with unique device mounting considerations, network cabinets. (1 day)
- Developed Sensor Installation Plan PDF (SIP) CAD Layouts and Visio Diagrams of networking for installable project by electricians. Device/Antenna locations, heights, pitches, rolls, yaws, cabling paths, switch ports, IP addresses, etc. Remote work (1 week)
- Deployed System. Networked devices and got communicating to server, IP to MAC registries with IT and any VLAN or other issues diagnosed. Other Network troubleshooting. Vendor Software Configuration. Integration file configurations. Program RTLS tags (update rate, status) and associate tags to assets. Integrated to in house JD automation software. Asset Tag to OPC tag associations.
- RTLS Tags on ladders, hoists, and GCI arms to prevent tractors from leaving station when any assets were present.
- Surveyed and programmed 200 UWB devices in 3 days in a metal-rich plant during live production using a manual total station. Devices surveyed to within 5mm for safety application.
- Project deployed and accepted in 2 weeks. 15 weeks total and 2 trips.
- Led outdoor GPS correction and deviation analysis. Lead correction effort with software team in Cambridge UK and reduced GPS error across different yards from 1–5 m.
- Digitized facility CAD into 3d and 2d models;
- Designed 3 additional projects while there totaling 300 devices on FEE lines that I later deployed.

GM Factory Zero RTLS Deployment**Hamtramck, Michigan**

- **Surveyed and programmed 1,000 UWB devices in one weeks with LiDAR and calibration software built with the Ubisense founders** in Cambridge, UK. Survey accurate to 2cm. Stitched lidar scan control points together with total station.
- Generated coordinate system with missing line measurements translated to columns in facility CAD files.
- Surveyed and calibrated devices simultaneously while walking around with SLAM Lidar equipment mounted with UWB tag.
- Tool-control project: Radar positioning tied to tool controllers on the EV line. Torque settings programming and logs for quality and target recalls. Sigma 6
- Trained teammates on revised lidar delivery technique, rented lidar scanners, and utilized new methodology to deploy additional 4000 sensors within same facility simultaneously within 2 weeks.
- Procured lidars and total stations and created pelican cased kits. Issued to teammates to facilitate on site travel.

Lockheed Martin F-35 & C130 Factory RTLS Deployment**Fort Worth, Texas**

- Surveyed and programmed 100 UWB devices across final assembly, storage buildings, flight line, building exteriors, and hangars in 1 week using manual total station. Surveyed during live production.
- UWB indoors and GPS outdoors. RTLS system accurate to 2 cm inside and 5 m outside.
- Deviation analysis in a local coordinate system. CAD/SVG maps and QA/QC so the floor system matched the drawings. Revised SVG webmaps for aggregated asset tracking application. Traveled with kit.
- FOD training and on base protocols

Daimler Sprinter Van Factory RTLS Deployment

Charleston, South Carolina

- Surveyed and programmed 300 UWB Devices Tool-control RTLS integrated with PLCs across assembly, EOL, and outdoor yards. Surveyed during live production in one week to 2cm with manual total station.
- Contracted external surveyors for parallel on site work to survey an additional 200 devices. Performed QA/QC on their work. Traveled with kit.

Mitchell Elementary School Solar Installation

Golden, Colorado

- Volunteered, designed, and managed a solar installation project for Mitchell Elementary.
- Collaborated with teachers and administrators to create a hands-on learning experience for students.
- Featured on local news for inspiring STEM awareness in the community

ADDITIONAL

Open to learning additional survey techniques such as utilizing a robotic total station, drones, or other survey equipment. Field Experience only. No PLS or CST at this time

INTERESTS

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| ▪ Community: Coaching, education, volunteering | ▪ Art: Photography, drawing, painting, web development, game development |
| ▪ Sports: Football, basketball, weightlifting, yoga | ▪ Reading: History, philosophy, spirituality, technology, science fiction |
| ▪ Outdoor Activities: Hiking, hiking | ▪ Geology: Mineral Survey, Prospecting, |

REFERENCES

Available upon request.